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Allometric Scaling Laws for Bipedal Robots

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***Abstract*— Scaling the design of robots up or down remains a fundamental challenge. While biological systems follow well-established isometric and allometric scaling laws relating mass, stride frequency, velocity, and torque, it is unclear how these relationships translate to robotic systems. In this paper, we generate similar allometric scaling laws for bipedal robots across three orders of magnitude in leg length. First, we conduct a review of legged robots from the literature and extract empirical relationships between leg length (L), body length, mass, and speed. These data show that robot mass scales more closely to L^2, in contrast to the L^3 scaling predicted by isometric scaling. We then perform controlled simulation studies in Drake using three variants of real quasi-passive, hip-actuated walkers with different foot geometries and control strategies. We evaluate the performance of each design scaled with leg length, L. Across all robots, walking velocity follows the expected $L^{\frac{1}{2}}$ trend from dynamic similarity. Minimum required torque scales more closely with mL than the isometric model of mL^2. Foot geometry scaled proportionally with L^1. These results provide new insight into how robot designs allometrically scale to different sizes, and how that scaling is different from isometric or biological scaling laws.**

I. Introduction

Designing effective legged robots is still a challenge in robotics, especially as we look to larger and smaller sizes. In principle, one would like to scale an existing robot to a target size while preserving its locomotion performance and stability. In practice, however, the appropriate scaling relationships and the extent to which performance transfers across scale are not well understood, making it difficult to predict how a scaled design will behave.

In biology, the characteristic properties of animals, such as mass, stride frequency, velocity, and torque, follow predictable scaling laws. These start with *isometric*, or *geometric*, scaling based on geometry and first principles—that all lengths scale proportionally, that mass scales proportionally to volume ($m \propto L^3$), etc. [1,2]. Any deviations from this relationship are called *allometric* scaling, including trends due to elastic similarity [3,4], dynamic similarity [1], constant stress [5], or empirical relationships [6]. But, it is unclear how well these relationships translate to and inform the design of robotic platforms at different scales since robots use different materials, actuators, controllers, etc. [7].

In prior work, researchers have tried to scale robots using isometric and allometric scaling inspired by biology [8–12].

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Fig. 1: Two of the robots considered in this paper, the 15cm leg length Mugatu [13] and the 2.5cm leg length Zippy [14].

One example, that inspires this paper, is that of the simple biped robot from [13], which was scaled down from 0.15 m to 0.025 m leg length in [14] (Figure 1). The authors of that work found that they could not simply scale the design down following isometric scaling. Instead, the smaller robot needed a different foot shape, but was relatively faster than the larger robot. This leads to the question, **is there something fundamental about size or scale that required these changes or resulted in these performance differences?**

More generally, the key research questions considered in this paper are: **How do robot designs change at different sizes? Do robots follow the same isometric and allometric scaling laws as biology?** And, **will a given robot design work better or worse as we make it larger or smaller?**

To address these questions, this paper makes the following contributions: 1) We derive new empirical scaling laws based on a survey of existing robot systems with different numbers of legs (Section II). 2) To eliminate some of the variation between robot designs and test scaling laws more precisely, we develop a scalable simulation framework of simple bipedal robots at a range of different leg length scales from 0.025 m to 1.0 m (Section III). 3) We then use this simulation to assess how velocity, torque, body attitude, and foot shape scale with a single design (Section IV). 4) Finally, we compare these new relationships to those determined from first principles and biology in order to understand how robot designs scale (Section V). The overall result is a new

	Isometric	Biology	Robotics
Body Length, $L_B \propto$	L^1	$L^{0.9}$	$L^{0.8}$
Body Mass, $m \propto$	L^3	$L^{2.4}$–L^3	L^2
Velocity, $v \propto$	$L^{0.5}$	$L^{0.25}$	$L^{0.5}$
Torque, $\tau \propto$	mL^2	L^3	mL

TABLE I: Allometric scaling laws for robotics based on leg length, L.

understanding of allometric scaling for robotics, summarized in Table I.

In particular, we focus on three surprising aspects of design. The first is mass, where we find that legged robot mass scales much slower than isometry predicts ($m \propto L^2$ instead of $m \propto L^3$), meaning that larger robots are relatively light and smaller robots are relatively heavy. The second aspect relates to torque requirements, where we find that the torque demand for bipeds ($\tau \propto mL$) does not scale as quickly as previously thought ($\tau \propto mL^2$), and that it closely matches the torque capabilities of motors at different sizes. Finally, we analyze the physical shape of the feet, which are very important for robots like those in Figure 1, where we find that the differences in dynamics at different sizes (in part due to the different masses) means that robots need different body shapes at different size scales.

This study is timely, as these questions could not be as carefully addressed until recently. Legged robots have exploded in popularity in the last few years, leading to an increase in the number and diversity of designs. This is critical in order to have enough data at sufficiently different sizes to reliably discriminate between different scaling laws. Additionally, simulation environments have greatly improved, which is especially important for trying to simulate very small or very large robots.

A. Related Work

Isometric scaling with proportional length growth is generally followed in animals, but some key deviations appear. Mass generally scales isometrically with body length cubed, as $m \propto L_B^{3.00}$ [15] (with other studies ranging from $m \propto L_B^{2.80}$ [16] to $m \propto L_B^{3.18}$ [4])[1]. However, for limb or leg length, mass grows a little slower than the cube of limb length, or equivalently that limbs get relatively longer as the animal gets larger. For example, one study found that mammalian mass versus bone length scales as $m \propto L^{2.85}$ [6], or another found $m \propto L^{2.5}$ for forelimbs and $m \propto L^{2.7}$ for hindlimbs of quadrupedal mammals [17]. This allometry is even more evident in avian bipeds, which exhibit $m \propto L^{2.44}$ [18], suggesting that bipedalism may require different scaling than other forms of locomotion.

Beyond morphology, locomotor speed also exhibits systematic scale dependence. Past work demonstrated that maximum speed for reptiles, aves, and mammals follows a hump-shaped scaling law, explaining why the largest animals are not the fastest, due to acceleration limits and metabolic constraints [19,20]. Before that peak, velocity scales as $v \propto L^{0.25}$ (See Figure 1 [20] and data within).

Recent cross-domain analyses reinforce an ambiguity in torque and power. Under isometric similarity, rotational arguments predict joint torque demands on the order of $\tau \propto mL^2 \propto L^5$. However, both biological and robotic systems routinely deviate from this upper-bound scaling. Models of torque profiles have been used to understand scale-dependent active muscle responses in biological hip joints, showing that torque demand appears to scale as only $\tau \propto L^3$ [21]. Some analyses extend beyond the biological muscle power, body mass, and acceleration time, to a general power-mass-speed relationship: in [22,23] they find that both animals and ground-mobile machines obey a common power-mass-speed scaling law, suggesting deep physical constraints that transcend implementation.

In robotics, however, scaling laws are less well established and are not limited by evolutionary, ecological, and biological constraints. Isometric scaling, since it is based on first principles and geometry, still provides a good starting point in engineering [24]. It is commonly used to scale robotic systems, but does not capture allometric effects. Just as in biology, the different rates at which properties scale and the different materials and components available at different sizes mean that pure isometric scaling does not capture how robots actually differ across sizes. For example, [11] found that elastic similarity was a better model of how their small quadrupeds scaled.

As for how motors scale, there is a disagreement in the literature. Theoretical models for torque relative to motor diameter range from $\tau \propto L^{3.5}$ [25,26], to $\tau \propto L^4$ [27–29], and to as high as $\tau \propto L^5$ [24,25], depending on assumptions about magnetic loading, current density, and cooling. Empirical studies span a similar range, including $\tau \propto L^3$ [30], $\tau \propto L^{3.5}$ [26,31], $\tau \propto L^{3.8}$ [30,32,33], $\tau \propto L^4$ [28,34], and as high as $\tau \propto L^5$ [24]. Beyond torque, the stress-limited speed appears to scale as $\omega \propto L^{-1}$, constraining the joint power envelope at large sizes [24,25,28,31]. The scaling of the gear-box and transmission also plays an important role [35].

II. Scaling Trends in Existing Robotics

In this section, we analyze the trends in existing robot designs. We collected data on robots spanning bipedal, quadrupedal, and hexapedal morphologies, and recorded each platform's leg length, overall body length, mass, nominal walking or running speed, and degrees of freedom (DOF).

Bipedal robots were split into two groups based on the morphology and degrees of freedom. Bipedal robots designed without an upper torso or with less than 4 DOFs were sorted into the "lower bipedal" category. Other bipedal robots were classified as "full bipedal" robots. In total, we found data for 10 lower bipeds, 17 bipeds, 14 quadrupeds, and 6 hexapods. The full data table, including sources, is in an open source repository. The regression results are summarized in Table II and Figure 2.

[1]Note that in the biology literature, scaling is generally done with respect to mass (i.e. $L \propto m^{1/3}$), however in terms of robot design using length as the base is more applicable. Therefore, for consistency in this paper, we have presented all scaling laws with respect to length.

	Lower Bipedal Robot	Full Bipedal Robot	Quadrupedal Robot	Hexapedal Robot	Global
Number in group $n =$	10	17	14	6	47
Body Length [m] $L_B \propto$	$L^{1.00}$ (R^2=0.985)	$L^{0.87}$ (R^2=0.728)	$L^{0.81}$ (R^2=0.872)	$L^{0.79}$ (R^2=0.980)	$L^{0.86}$ (R^2 = 0.865)
Body Mass [kg] $m \propto$	$L^{1.95}$ (R^2=0.962)	$L^{1.91}$ (R^2=0.560)	$L^{2.72}$ (R^2=0.928)	$L^{2.23}$ (R^2=0.881)	$L^{2.12}$ (R^2 = 0.867)
Body Mass [kg] $m \propto$	$L_B^{1.96}$ (R^2 = 0.984)	$L_B^{2.27}$ (R^2 = 0.819)	$L_B^{3.14}$ (R^2 = 0.942)	$L_B^{2.89}$ (R^2 = 0.950)	$L_B^{2.34}$ (R^2 = 0.902)

TABLE II: Results from survey of existing robot designs.

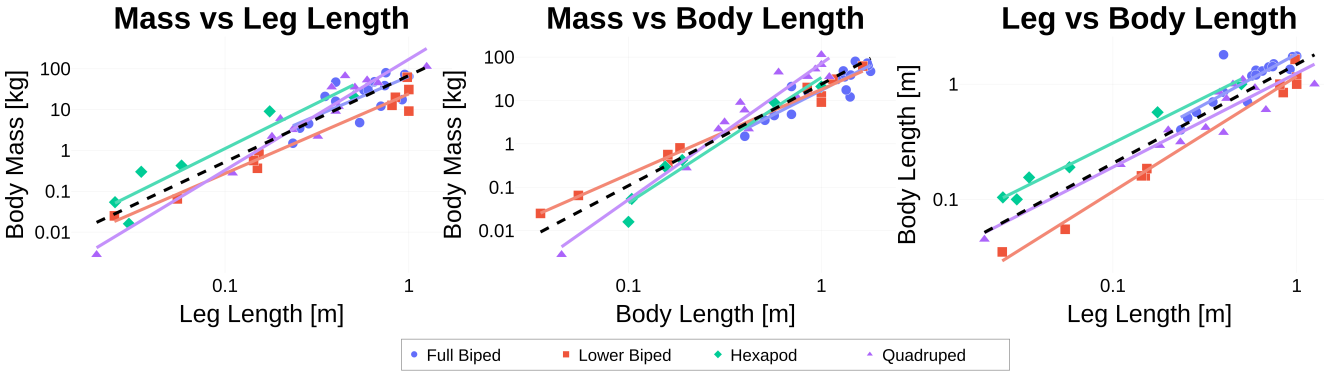


Fig. 2: Relationships between body length, leg length, and body mass across multiple existing robot morphologies.

Leg length is defined as the robot's hip height in a nominal standing posture. Body length is defined as the robot's longest dimension in a nominal standing posture, generally height for bipeds and length for quadrupeds and hexapods. For all of the robots except the lower bipedal robots, we find that the body length L_B scales slower than the leg length L, about $L_B \propto L^{0.79}$, showing a divergence from the relationships assumed from isometric scaling ($L_B \propto L^1$). On average, as robots get bigger, their legs become relatively larger compared to their body. In contrast, lower bipedal robots, which are almost only legs, followed the isometric relationship of $L_B \propto L^{1.00}$.

Looking at mass versus leg or body length, we find the surprising relationship of approximately $m \propto L^2$, lower than isometric scaling, $m \propto L^3$. Between the different types, we find that the bipeds and the multi-legged robots follow somewhat different trends. The quadrupedal and hexapedal robots' mass scales a little faster than L^2 compared to leg length and more closely follow the isometric scaling prediction of $m \propto L_B^3$ when comparing mass to body length. However, the bipedal robots (both lower and full) show a lower mass scaling with leg length of just below $m \propto L^2$.

Additionally, we compared leg length and body length to forward velocity. We were unable to observe statistically significant trends in these relationships. We attribute this to inconsistent reporting in forward velocity. While some works specify walking, running, or trotting, others do not clearly describe the gait. Some works report average velocity while others report peak velocity. Finally, many robots are optimized for payload capacity or energetic efficiency and do not try to determine peak velocity.

III. Simulation and Analysis Methods

While we were able to find surprising scaling trends across designs in the previous section, the vast design differences between different examples makes further analysis difficult. In this section, we consider the exact scaling of specific designs in simulation in order to analyze other properties of speed, torque, and foot shape for quasi-passive biped walkers. Further studies are required to see if these trends generalize to other designs.

To make this direct comparison, we simulate a series of bipedal robots in a 3-D simulator, Drake [36]. This simulator was chosen because of its hydroelastic contact model, which allows high-fidelity contact modeling for systems like passive dynamic walkers that are sensitive to initial conditions and have rolling contact. To ensure that the simulator was able to capture the behaviors of the real hardware, simulations at the $1\times$ scale were compared to experimental results on hardware. The simulator was set up to use the same control scheme, mass distribution, body shape, and actuation parameters (i.e. sinusoidal control vs bang-bang control, motor torque-speed relationship) as the real hardware. The hydroelastic contact parameters were tuned until the body roll, pitch, and yaw, matched those achieved by the real hardware with the hardware's actuation parameters. Achieving reliable results in simulation is difficult, particularly at small scales due to friction, inertial, and contact forces. To ensure matching at smaller scales, the mesh resolution parameter was decreased proportionally in accordance with Drake guidelines [36].

The study focuses on two robot designs (Figures 1 and 3), which we call Zippy and Mugatu, that share the same general morphology but differ in actuation strategy, foot shape, and mass distribution [13,14]. Both robots are quasi-passive biped walkers with a single actuator at the hip and curved feet, enabling passive dynamics to contribute to forward

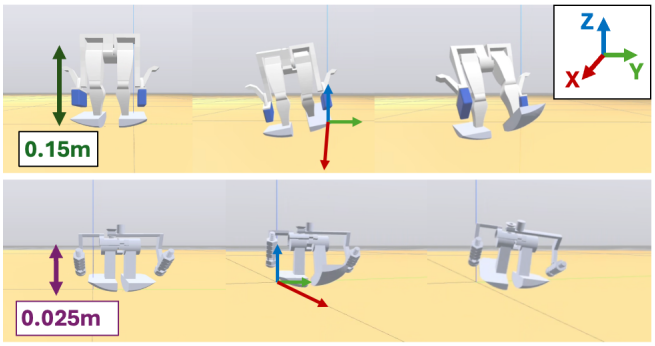


Fig. 3: Mugatu (Top) and Zippy (Bottom) robots in the Drake simulation (different scales for clarity).

	Mugatu	**Zippy**	**Zippy**
Foot Shape	Spherical	Ellipsoidal	Spherical
Leg Length [mm] L	153.0	24.7	24.3
Body Length [mm] L_B	186.0	36.4	36.4
X,Z Semi-Axis [mm] R_X,R_Z	120.0	25.0	19.1
Y Semi-Axis [mm] R_Y	120.0	30.0	19.1
Body Mass [g] m	900.0	25.0	25.0

TABLE III: Physical parameters of robot configurations tested in simulation (estimated from CAD).

motion. This morphology is a good candidate for this study as it eliminates unnecessary complexity – there is only one actuator and therefore fewer control parameters to consider. While they generally have the same morphology, Zippy uses ellipsoidal feet and a bang-bang torque control scheme with mechanical hardstops, while Mugatu employs spherical feet and a sinusoidal position-controlled input at the hip. To try and distinguish between these differences, we also consider a third simulation model of the Zippy robot but with spherical feet (i.e. scaled Mugatu feet, Table III).

CAD models of each robot were obtained from the authors of [13,14]. Each variation was simulated at a range of scales between $L = 0.02\,\text{m}$ to $1.2\,\text{m}$, including points at the leg lengths of Zippy ($0.025\,\text{m}$) and Mugatu ($0.15\,\text{m}$), as well as an imagined large human sized robot at 1.0 m. When scaling these designs in simulation, all lengths were scaled equally, matching the relationship found for lower bipedal robots in the previous section. Mass was scaled with both mass scaling theories ($m \propto L^2$ and $m \propto L^3$), stride frequency was scaled as $f \propto L^{-1/2}$ to maintain dynamic similarity, and the minimum torque required to achieve consistent walking was manually tuned.

For each condition, the robot's locomotion is evaluated after stable, consistent walking was achieved, defined as the time at which the phase plot of body pitch and pitch rate converged to a 2-D limit cycle. The following outputs were measured for successful trials:

1) Walking velocity
2) Torque at the hip joint
3) Body attitude: center of mass roll, pitch, and yaw

To determine the amplitudes of body attitude oscillation, roll, pitch, and yaw were measured for the duration of the walking trial. For each body attitude metric, the peaks and valleys of the signal over the course of each walking cycle were determined. These values were subtracted and averaged to determine the average peak-to-peak amplitudes.

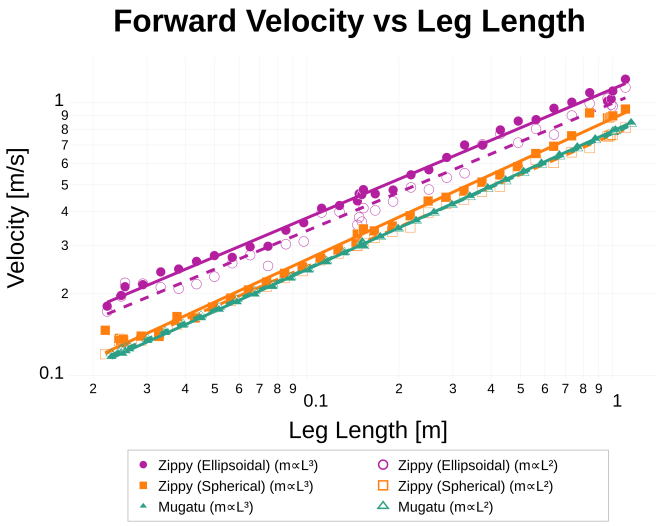


Fig. 4: Simulated forward velocity of robots across scales (see Table IV).

Robot design is further evaluated through a controlled study of foot-curvature scaling. Both robots are compared at the three characteristic body scales with each robot's foot shape independently scaled along the X, Y, and Z axes from $0.5\times$ to $2.0\times$ in increments of 0.1. For every resulting foot-shape variation, the robot is simulated for 25.0 s. During each simulation, body attitude, body position, and other body-state data are recorded. These data are then analyzed to identify trends in body dynamics and locomotion.

IV. Simulation Results

A. Mass, Frequency, and Velocity Scaling

The results for the robot velocity as a function of mass and scale are shown in Figure 4 and Table IV. Across scales and all three robot morphologies, walking velocity increased with leg length, closely following the dynamic similarity relationship $v \propto \sqrt{L}$. In particular, the two mass scaling laws behaved similarly, suggesting that the walking speed primarily depends on leg length and not mass. This means that each robot design walked with the same Froude number at each size scale [1]. The walking strategy of these robots, that relies on passive dynamics, makes them closely tied to dynamic similarity, while robots with other walking strategies may deviate.

While following similar scaling laws, the absolute values of the walking velocities were not the same across morphologies. Zippy with ellipsoidal feet walked faster than both Zippy with spherical feet and Mugatu across all sizes. Zippy with spherical feet and Mugatu perform more closely, which is likely a result of the similar foot curvature.

	Zippy (Ellipsoidal)	Zippy (Spherical)	Mugatu
	$m \propto L^2$		
Velocity [$\frac{\text{m}}{\text{s}}$] $v \propto$	$L^{0.47}$ (R^2=0.979)	$L^{0.48}$ (R^2=0.998)	$L^{0.50}$ (R^2=1.000)
Minimum torque [N m] $\tau \propto$	$L^{2.92}$ (R^2=0.998)	$L^{2.93}$ (R^2=0.999)	$L^{3.49}$ (R^2=0.974)
Roll amplitude [°] $\theta_R =$	9.28 ± 0.59	16.27 ± 0.47	19.91 ± 0.32
Pitch amplitude [°] $\theta_P =$	25.02 ± 3.05	25.02 ± 1.89	10.83 ± 0.13
Yaw amplitude [°] $\theta_Y =$	7.14 ± 1.12	7.45 ± 0.93	10.06 ± 0.30
	$m \propto L^3$		
Velocity [$\frac{\text{m}}{\text{s}}$] $v \propto$	$L^{0.47}$ (R^2=0.994)	$L^{0.52}$ (R^2=0.994)	$L^{0.51}$ (R^2=1.000)
Minimum torque [N m] $\tau \propto$	$L^{3.90}$ (R^2=0.999)	$L^{3.96}$ (R^2=0.999)	$L^{4.07}$ (R^2=0.981)
Roll amplitude [°] $\theta_R =$	10.09 ± 1.23	16.60 ± 0.41	19.91 ± 0.32
Pitch amplitude [°] $\theta_P =$	27.69 ± 3.07	29.22 ± 4.86	10.82 ± 0.24
Yaw amplitude [°] $\theta_Y =$	8.43 ± 2.21	7.82 ± 1.01	9.82 ± 0.40
	Feasible Foot Curvatures		
X Semi-Axis Length [m] $R_X \propto$	$1.8L^{1.01}$ (R^2=0.978)	—	$1.2L^{1.00}$ (R^2=0.978)
Y Semi-Axis Length [m] $R_Y \propto$	$1.4L^{1.01}$ (R^2=0.957)	—	$1.1L^{0.99}$ (R^2=0.965)
Z Semi-Axis Length [m] $R_Z \propto$	$0.99L^{0.99}$ (R^2=0.930)	—	$0.78L^{1.00}$ (R^2=0.929)

TABLE IV: Simulation results for velocity, torque, body attitude, and foot shape scaling.

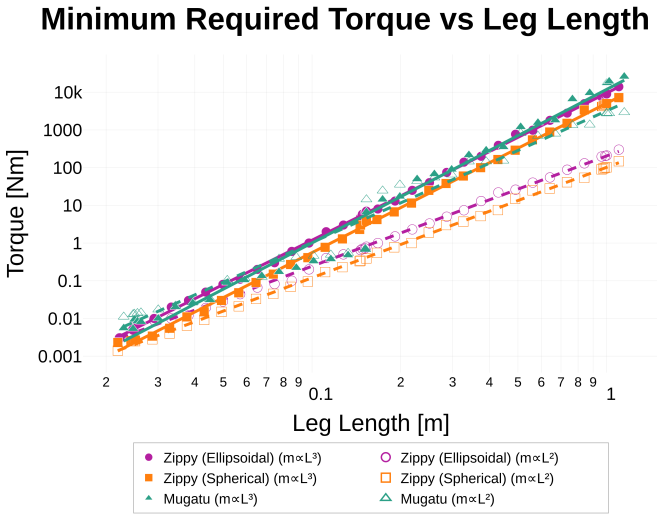


Fig. 5: Simulated minimum required torque of Zippy and Mugatu across scales (see Table IV).

B. Torque Scaling

Next, we considered how the torque required to walk changes with size (Figure 5 and Table IV). For $m \propto L^2$ scaling, the minimum torque closely follows the $\tau \propto L^3$ trend predicted by biological analysis (with Mugatu trending a little higher). For $m \propto L^3$ scaling, all robots showed a torque scaling closer to $\tau \propto L^4$. Combining these results, we see an overall scaling of approximately $\tau \propto mL$.

In all cases, the robots require less torque than that predicted by geometric similarity of $\tau \propto L^5$ or mL^2 [24]. Furthermore, the required torque more closely follows the $\tau \propto L^{3.5}$–L^4 trend seen in available motor torque, suggesting these designs will work well across a range of sizes.

C. Foot Scaling and Body Attitude

1) Isometric Foot Scaling: When the entire foot was scaled isometrically ($m \propto L^2$ and $m \propto L^3$), the roll, pitch, and yaw amplitudes remained broadly consistent across scales for each robot design, with no strong systematic trend as a function of leg length (Table IV). This suggests consistent dynamics across scales.

2) Feasible Foot Geometries: The analysis of foot axis length versus leg length reveals that successful foot axis lengths scale with body size following a clear power-law relationship across all three axes $L^{0.99} - L^{1.01}$. This linear scaling indicates that foot axis length should grow proportionally with leg length, which is consistent with geometric similarity, and suggests the range of feasible morphologies is dependent on initial robot morphology (see similar slopes but different y-intercepts in Figure 6).

The foot axis lengths that yield stable walking are consistently larger for Zippy than for Mugatu across all leg-length scales. Larger axis lengths correspond to lower foot curvature (i.e. flatter feet), so Zippy requires flatter feet at any given leg length to walk stably, potentially due to differences in the inertia of its other components.

3) Optimal Foot Geometries: Within the range of feasible foot shapes, different foot shapes produce the fastest walking for each robot at each scale. At the largest scale ($L = 1.0\,\text{m}$), Mugatu achieves a slightly higher peak velocity achieved by the optimal foot design than Zippy ($1.84\,\frac{\text{m}}{\text{s}}$ vs. $1.74\,\frac{\text{m}}{\text{s}}$) (Figure 7, left). In contrast, Zippy at the smallest scale ($L = 0.025\,\text{m}$) reaches a higher peak velocity than Mugatu ($0.253\,\frac{\text{m}}{\text{s}}$ vs. $0.228\,\frac{\text{m}}{\text{s}}$).

Next, we compare the peak velocity to the predicted velocity based on scaling the robot from the original $1\times$ scale by dynamic similarity, or Froude (Figure 7, center). The progression for both robots roughly follows Froude scaling ($v \propto \sqrt{L}$) between the small and mid scales, but there are deviations. Mugatu exceeds Froude predictions by 24 % at $L = 1.0\,\text{m}$ (comparing $L = 0.15\,\text{m}$ and $L = 1.0\,\text{m}$, expected: 2.56, observed: 3.17). The observed velocity ratio of Zippy when simulated is 6.87, while the expected is 6.32 (comparing $L = 0.025\,\text{m}$ and $L = 1.0\,\text{m}$), exceeding the velocity predicted by Froude number and dynamic similarity by roughly 9 %.

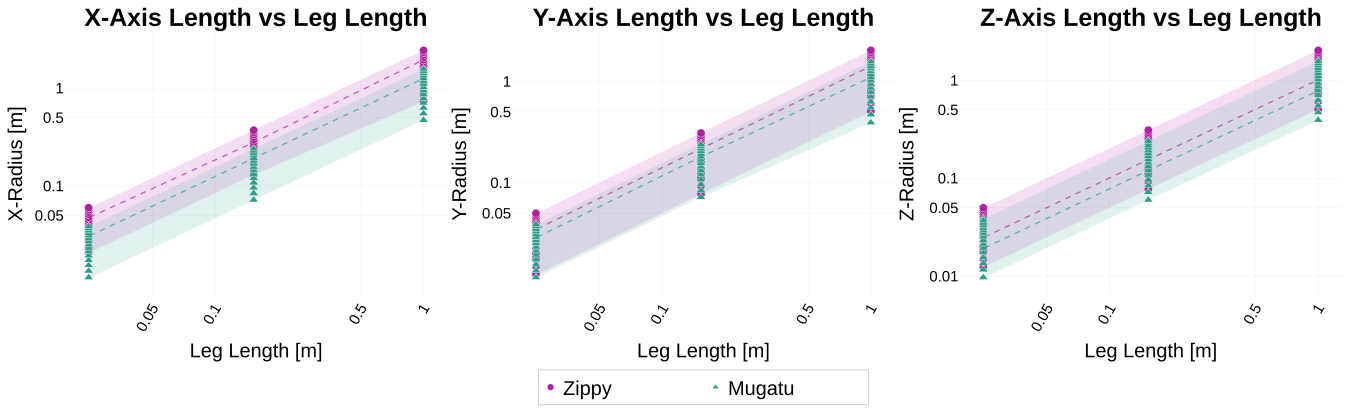


Fig. 6: X, Y, Z axis lengths that resulted in stable walking in Mugatu and Zippy.

Mugatu velocity thus exhibits greater-than-Froude scaling at the largest leg length, overtaking Zippy's absolute velocity. This departure suggests that optimized foot geometry at larger scales provides additional dynamic stability beyond what pure geometric scaling would predict, and that Mugatu's morphology captures this benefit more effectively than Zippy's.

We also considered the aspect ratios of the foot shape that achieved the highest velocity at each scale (Figure 7, right). The aspect ratio that achieved the highest velocity for each robot at each size was different and did not follow any clear trend. This difference in behavior is likely due to differences in body inertia between the two robots.

V. Discussion

The results of this study reveal that the scaling behavior of bipedal robots diverges from classical isometric predictions in several important ways, and that these deviations are consistent across robot morphologies and control strategies. Here, we contextualize these findings with respect to biological scaling theory, discuss practical implications for robot design, and acknowledge limitations of the current approach.

A. Mass Scaling

We see deviations between isometric, biological, and robotic trends with strong correlations (Table I). Perhaps the most striking finding from the survey of existing robots is that bipedal robot mass scales approximately as $m \propto L^2$, well below the $m \propto L^3$ predicted by isometric scaling and observed in most biological systems. This result implies that smaller bipedal robots are relatively heavier for their size, while larger robots are relatively lighter. This trend can be attributed to the fact that robot designers do not simply scale all components uniformly: electronics, fasteners, actuators, and structural elements each follow their own size-availability and manufacturing constraints. At small scales, the mass of motors, batteries, and circuit boards constitutes a larger fraction of total body mass than it would at larger scales, where structural mass dominates. In contrast, quadrupedal and hexapedal robots in the survey followed mass scaling closer to $m \propto L^3$, suggesting that the additional legs and generally larger bodies bring these platforms closer to volumetric scaling. The distinction between bipedal and multi-legged mass scaling, which mirrors some differences in biology [18], has direct consequences for torque requirements, stability margins, and energy efficiency, and should be accounted for in any principled approach to scaling robot designs.

B. Velocity Scaling and Dynamic Similarity

The walking velocity achieved across robot morphology, scale, and scaling theory ($m \propto L^3$ and $m \propto L^2$), still followed the trends predicted by the dynamic similarity hypothesis derived from geometric similarity. This means that the robots maintained a roughly constant Froude number across scales, something seen in dynamically similar gaits in both biology and robotics. Importantly, the two mass scaling laws ($m \propto L^3$ and $m \propto L^2$) produced nearly identical velocity trends, indicating that walking speed in these quasi-passive dynamic walkers is primarily governed by leg length and pendular dynamics rather than by mass. This is consistent with the physics of passive walking, where gravitational potential energy and leg geometry set the natural gait speed rather than mass.

However, while the scaling exponent was consistent, the absolute velocities differed across morphologies. Zippy with ellipsoidal feet was consistently faster than both Zippy with spherical feet and Mugatu at every scale. This suggests that foot geometry, body inertia, and actuation strategy, not just scale, play a role in determining locomotion performance. The similarity in velocity between Zippy with spherical feet and Mugatu points to foot curvature as a key differentiator, while the additional speed advantage of the ellipsoidal-footed design may also reflect the contribution of the body inertia and bang-bang control scheme with mechanical hardstops, which introduce impulsive energy injection during the gait cycle.

C. Torque Demand Scaling

The torque analysis provides an encouraging result for the scalability of legged robots. Under $m \propto L^2$ mass scaling,

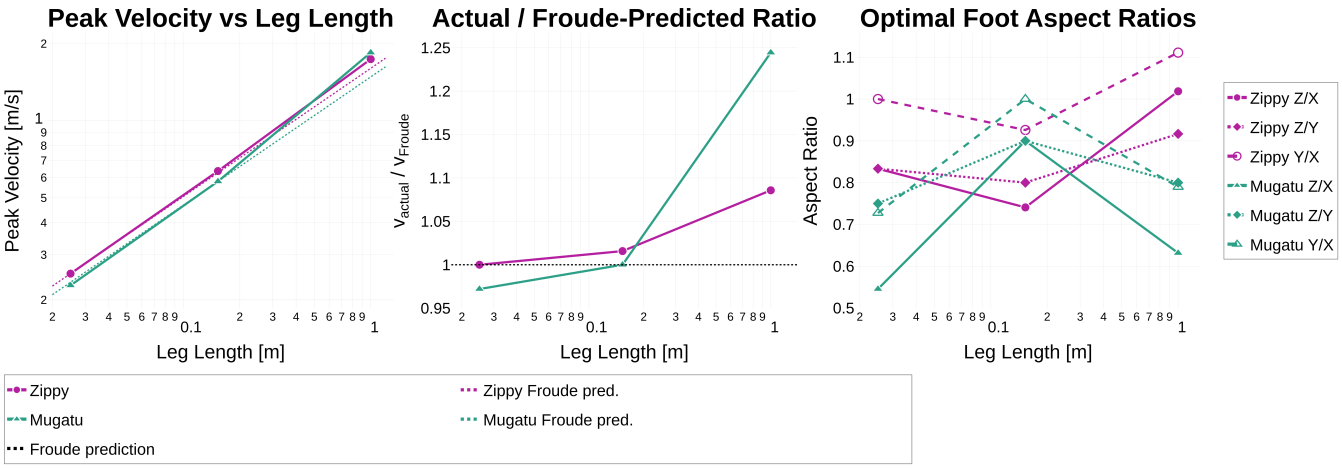


Fig. 7: Peak velocity, found by optimizing along the X, Y, Z axis, across scales in Mugatu and Zippy. This is compared to Froude predicted speed across scales. Ratios of the foot: The $\frac{Y}{X}$ foot width vs foot length (transverse plane), the $\frac{Z}{X}$ foot height vs foot length (sagittal plane), and the $\frac{Z}{Y}$ foot height vs foot width (frontal plane).

the minimum torque required for walking scaled as approximately $\tau \propto L^3$, while under $m \propto L^3$ scaling, it rose to approximately $\tau \propto L^4$. In both cases, the combined relationship simplifies to $\tau \propto mL$, which is substantially lower than the $\tau \propto mL^2 \propto L^5$ predicted by isometry. This reduced torque demand is important because empirical surveys of available motor torque suggest a scaling in the range of $\tau \propto L^{3.5}$ to $\tau \propto L^4$, meaning that the torque requirements of these passive-dynamic walkers fall within or below the capabilities of commercially available actuators across a broad range of sizes.

This finding is also consistent with biological observations, where hip torque in animals scales as roughly $\tau \propto L^3$ rather than the isometric $\tau \propto L^5$, reflecting the role of posture, gait optimization, compliance, and muscle force scaling in reducing joint loading. The implication for robot design is that designers need not assume worst-case isometric torque budgets when selecting actuators for scaled platforms; doing so would lead to over-engineered, unnecessarily heavy, and less efficient designs.

D. Foot Shape Scaling and Body Attitude

The foot curvature study reveals that while the axis lengths enabling stable walking scale nearly linearly with leg length, consistent with geometric similarity, the optimal foot shape for maximizing velocity does not remain constant across scales. This means that a foot shape optimized at one scale will generally not be optimal at another, even if the overall foot size is scaled proportionally.

This result has practical implications: it suggests that foot design should be treated as a scale-dependent optimization problem rather than a fixed geometric ratio. This is likely due to an interplay between body inertia, contact dynamics, and gravitational loading, all of which change at different rates with scale.

In comparison, body attitude across morphologies remains fairly consistent. However, notable differences exist between the robot designs themselves: Mugatu exhibited lower pitch amplitudes but higher roll and yaw amplitudes than Zippy, reflecting differences in mass distribution and foot geometry. These design-specific offsets underscore the point that while scaling laws capture general trends, the specific performance envelope of a robot is shaped by its morphological details.

VI. Conclusions and Future Work

This work addresses scaling laws in the quasi-passive robotic systems Zippy and Mugatu, finding surprising new relationships between body mass, leg length, and required torque.

Several limitations of this study should be acknowledged. First, the simulation framework, while calibrated against physical hardware, does not capture real-world effects such as manufacturing limitations, computational limitations, and friction variability which become increasingly relevant at small scales and more complex designs. Second, the robots considered are quasi-passive walkers with a single hip actuator, representing a narrow slice of the design space for legged robots. Third, the empirical survey of existing robots, while spanning three orders of magnitude in leg length, is limited in sample size, particularly for hexapods and lower bipeds, and is subject to inconsistencies in how different research groups report performance metrics such as speed.

Future work could extend these simulated analyses to full bipedal robots and to quadrupedal platforms, and should investigate the scaling of power and energetic cost of transport, which is critical for practical deployment. Additionally, incorporating the scaling of transmissions, sensing, and control hardware into the analysis would provide a more complete picture of how total system performance scales. Finally, experimental validation at additional scales, particularly at the extremes of the size range would strengthen the conclusions drawn here.

In thinking about scaling up the simple biped design to $L = 1.0\,\text{m}$, based on the scaling laws for lower bipedal robots

($m \propto L^2$), the mass of a 1.0 m version should be approximately 33 kg rather than 1600 kg ($m \propto L^3$), which is much more achievable. Furthermore, the torque required scales favorably, and so we should be able to find a suitable actuator for this design. This new work could assess design and inertial differences between Mugatu and Zippy in controlled 3D simulations. These differences can include center of mass position, actuation strategy, actuator type, and inertia across the three morphologies.

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